

# LoCoBENCH: EVALUATING FACTUALITY AND LOGICAL COHERENCE IN LARGE LANGUAGE MODELS

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## ABSTRACT

Long-form factuality assessment decomposes a response into atomic claims and verifies each against retrieved evidence, treating the claims as independent. It therefore cannot see a response whose atoms are individually true but which does not hang together as an argument, because a later sentence retracts an earlier one or because the inferential links it draws do not hold. We formalize *logical coherence* as a property of the joint distribution over atom truth values induced by a Markov random field whose pairwise factors are the relations the response itself draws. The relation vocabulary is a two-level taxonomy: twelve interpretable discourse senses, grounded in PDTB 3.0 and RST, compiling deterministically onto five inferential couplings, each of which fixes a factor table by naming the world it penalizes. We define four coherence readouts, give the auxiliary-variable construction each requires, and argue for the mean posterior marginal on five stated criteria. Atom priors come from an existing factuality pipeline, making coherence and factuality two stages over one decomposition rather than competing scores. Validating such a measure needs data that varies coherence while holding content fixed, which no benchmark provides. We therefore introduce **LoCoBench**, a generator emitting *coherence ladders* — families of five responses over one claim set whose relative ordering is declared in advance — with an evaluation protocol scoring a measure on whether it recovers those orderings. Gold relation graphs satisfy 78.6% of the declared constraints; graphs mined from the response text reach 56.3% coupling-level F1 and 53.6% of the constraints, locating the bottleneck in relation extraction rather than the probabilistic model. We release the taxonomy, the generator, and the harness.

## 1 INTRODUCTION

The dominant paradigm for assessing the factuality of long-form generation is decompose-then-verify: split a response into atomic claims, check each against retrieved evidence, and aggregate the verdicts (Min et al., 2023; Wei et al., 2024; Song et al., 2024). The paradigm is effective and now standard. It also embeds an assumption — that the truth of each atom can be settled in isolation — which discards exactly the structure that makes prose an *argument* rather than a list.

Two failure modes escape it. A response can assert atoms that are individually supported but mutually inconsistent: a report stating both that no one was harmed and that three passengers died contains a true claim and a false one, and per-atom verification says so, but it does not register that the response contradicts *itself*. And the inferential links a response draws may not hold even when their endpoints do; a chain of true statements joined by a spurious “therefore” is not a sound argument. Xie et al. (2026) quantify the gap: on logic-related samples, SAFE, VeriScore and FactCheck-GPT omit the logical dimension on 60%, 78% and 48% of cases respectively.

**Factuality and coherence are orthogonal axes.** A response can be a bag of true facts that is incoherent, and it can be internally impeccable but wrong. Our running example (§10) is a recall report carrying an unresolved conflict; a second fixture pair summarizes one appellate judgment twice from the *same* facts, differing only in the order the claims are asserted, and only one is coherent. No amount of per-atom grounding separates them.

This paper measures the second axis, taking coherence to be a statement about a *joint* distribution: given what a response asserts and how it links its assertions, how much of it does the argument, taken as a whole, uphold? That question is answered by a marginal in a graphical model whose variables are atom truth values and whose factors are the asserted relations — not by a count of violated rules.

Our contributions are: (i) a **coherence MRF with a formal two-level relation taxonomy** (§4) — twelve discourse senses compiling deterministically onto five inferential couplings, each defined by the assignment it penalizes, which keeps the annotation vocabulary rich while the model stays a closed three-parameter pairwise family; (ii) **four readouts with a selection argument** (§5), each with its auxiliary-variable construction, showing the mean posterior marginal to be the only one monotone in conflict resolution, constant-free and bounded, and why two natural alternatives fail — one by scoring a disconnected bag of atoms perfectly; (iii) **two-stage composition** with factuality (§6); and (iv) **LoCoBench** (§7), a generator of coherence ladders with committee validation, an evaluation protocol and baselines (§8), and **measurements** (§9) separating extraction error from scoring error by swapping only the relation graph.

## 2 RELATED WORK

**Atomic factuality.** FactScore (Min et al., 2023) introduced per-atom precision against a knowledge source; SAFE (Wei et al., 2024) added search-augmented verification and  $F_1@K$ ; VeriScore (Song et al., 2024) restricted attention to verifiable claims. FactReasoner (Marinescu et al., 2025) replaced verdict counting with probabilistic inference over a factor graph of atom and context variables with NLI-derived pairwise factors. We reuse its decomposition, relation extractor and inference stack, and take its marginals as priors (§6); coherence is a different question asked of the same decomposition.

**The logic gap.** Xie et al. (2026) is the closest prior art and states the problem: decompose-then-verify pipelines verify atoms individually and overlook the logical dependencies among them, so a text combining true facts through a false dependency is misjudged as factual. Our departure is to model those dependencies as factors in a joint distribution and read coherence off its marginals, rather than treating them as a labelling task.

**Discourse coherence.** Entity-grid models (Barzilay & Lapata, 2008) and neural successors (Li & Jurafsky, 2017) score coherence from surface transition statistics, trained against synthetic perturbations — shuffling and contradiction injection. LoCoBench extends that methodology by making the perturbation an *operator on a declared relation graph* rather than on raw text, so its intended effect on the score is known per rung. DiscoScore (Zhao et al., 2023) uses BERT-based discourse representations. Our senses come from PDTB 3.0 (Prasad et al., 2019) and RST (Mann & Thompson, 1988), but as an interpretable layer compiling to a smaller inferential vocabulary, which is what makes them usable as factor types.

**Consistency checking, judges, and relational models.** SelfCheckGPT (Manakul et al., 2023) detects hallucination via pairwise NLI over sampled responses; its NLI variant is the natural ablation isolating what global inference adds over local detection. G-Eval (Liu et al., 2023) is the practical incumbent — an LLM judge scoring coherence directly. Markov logic (Richardson & Domingos, 2006) is the natural alternative encoding; Appendix H shows a naive one-clause-per-relation encoding is *not* equivalent to our MRF and is degenerate at its mode, while a three-clause expansion reproduces it exactly on the three-coupling fragment.

## 3 STAGE 1: ATOM PROBABILITIES

We take atom-level factuality probabilities from FactReasoner (Marinescu et al., 2025); we summarize it because the coherence model consumes its output and reuses its machinery (Figure 1). A response  $y$  is decomposed into atomic claims  $\mathcal{A} = \{a_1, \dots, a_n\}$ , each decontextualized so that it is independently checkable, and a retriever returns contexts  $\mathcal{C} = \{c_1, \dots, c_m\}$ . An LLM-based NLI extractor labels each admitted pair with  $r \in \{\text{entailment, contradiction, neutral}\}$  and a confidence  $p^*$ ; context–context pairs are scored both ways and entailment in both directions is reconciled to *equivalence*. Neutral verdicts are dropped and contribute no factor, leaving the atom at its prior. The confidence is read off the generation itself,

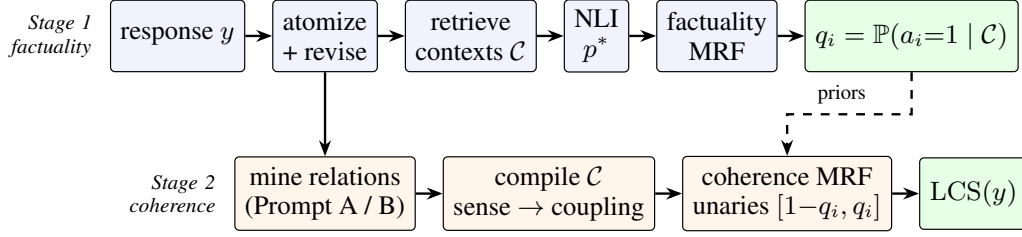


Figure 1: The two-stage pipeline. Stage 1 (§3) is FactReasoner (Marinescu et al., 2025): it grounds each atom against retrieved evidence and returns a posterior marginal  $q_i$ . Stage 2 (§4) builds a second MRF over the atoms *alone*, takes  $q_i$  as its unary priors, and adds one pairwise factor per mined atom–atom relation. Atomization is shared, so the response is decomposed once. Setting  $q_i \equiv \frac{1}{2}$  recovers a pure coherence score that consults no external evidence.

$$p^* = \exp\left(\frac{1}{|T|} \sum_{t \in T} \log p_t\right), \quad T = \{\text{tokens overlapping the emitted label span}\}, \quad (1)$$

the per-token geometric-mean probability of the label the model actually produced. Measuring exactly the span the label parser reports makes label and probability structurally unable to disagree, which matters because bracket delimiters are frequently fused into subword tokens. Where a backend exposes no logprobs,  $p^*$  comes from SIMBA-UQ (Bhattacharjya et al., 2025) instead.

The graph becomes a pairwise MRF — one binary variable per atom and per context, unary factors  $\phi_X(x) = [1 - \pi_X, \pi_X]$ , one pairwise factor per relation from Table 1 — and marginals are computed by weighted mini-bucket elimination (Liu & Ihler, 2011) with i-bound 6. Of its three graph variants we use FR2, which deduplicates contexts and scores every atom against all of them. We report the with-priors tables with  $\pi_C = 0.9$ ,  $\pi_A = 0.5$ ; the published model uses the no-priors tables and  $\pi_C = 0.99$ . The difference is immaterial to the coherence results, which set atom priors explicitly, but the tables are not interchangeable (§4.3).

## 4 STAGE 2: THE COHERENCE MRF

### 4.1 THE MODEL

**Definition 1** (Coherence MRF). Let  $\mathcal{A} = \langle a_1, \dots, a_n \rangle$  be the atoms of a response in assertion order, and let each  $a_i$  be a binary variable with  $a_i = 1$  read as “the claim holds”. Let  $\mathcal{R} = \{(s_r, t_r, \tau_r, p_r)\}_{r=1}^{|\mathcal{R}|}$  be a set of relations, where  $s_r \neq t_r$  index atoms,  $\tau_r$  is an inferential coupling (Def. 2) and  $p_r \in [0, 1]$  is the relation’s weight. The coherence MRF induced by  $(\mathcal{A}, \mathcal{R})$  and priors  $\pi \in [0, 1]^n$  is

$$\mathbb{P}(\mathbf{a}) = \frac{1}{Z} \prod_{i=1}^n \phi_i(a_i) \prod_{r \in \mathcal{R}} \psi_{\tau_r, p_r}(a_{s_r}, a_{t_r}), \quad Z = \sum_{\mathbf{a} \in \{0,1\}^n} \prod_i \phi_i \prod_r \psi_r, \quad (2)$$

with  $\phi_i(a_i) = [1 - \pi_i, \pi_i]$  and  $\psi_{\tau, p}$  given by Table 1.

Two properties matter. The factor family is *closed*: a coupling is only a choice of which cells of a  $2 \times 2$  table are down-weighted, so no relation type can smuggle in extra expressiveness. And the model is order-sensitive *only* through the relations — atom order enters via which pairs are related and in which direction, never a positional term — so an order-invariant edit provably cannot move the score, which the CONTROL ladder of §7 tests empirically.

### 4.2 LEVEL 1: INFERENTIAL COUPLINGS

**Definition 2** (Inferential couplings). A coupling  $\tau$  is specified by the set of joint assignments  $(a_s, a_t)$  it penalizes. The vocabulary is

Table 1: Pairwise potentials  $\psi_{\tau,p}(a_s, a_t)$ , row-major over  $(0, 0), (0, 1), (1, 0), (1, 1)$ ;  $\pi_s$  is the source prior and bold marks the penalized cell(s) of Def. 2. *No-priors* is correct for atom-context factuality, where the source’s own truth is at stake; *with-priors* is what coherence requires and what Def. 1 uses.

| Coupling $\tau$ | No priors                            | With priors                         | Property   |
|-----------------|--------------------------------------|-------------------------------------|------------|
| ENTAILMENT      | $[p, p, \mathbf{1-p}, p]$            | $[1-\pi_s, \pi_s, \mathbf{1-p}, p]$ | asymmetric |
| CONTRADICTION   | $[p, p, p, \mathbf{1-p}]$            | $[1-\pi_s, \pi_s, p, \mathbf{1-p}]$ | asymmetric |
| EQUIVALENCE     | $[p, \mathbf{1-p}, \mathbf{1-p}, p]$ |                                     | symmetric  |
| EXCLUSIVE       | $[\mathbf{1-p}, p, p, \mathbf{1-p}]$ |                                     | symmetric  |
| CO_NECESSITY    | $[\mathbf{1-p}, p, p, p]$            | $[\mathbf{1-p}, \pi_s, \pi_s, p]$   | symmetric  |

| <i>Coupling</i> | <i>Reading</i>                                     | <i>Directed</i> | <i>Penalized world(s)</i> |
|-----------------|----------------------------------------------------|-----------------|---------------------------|
| ENTAILMENT      | $a_s \Rightarrow a_t$                              | yes             | (1, 0)                    |
| CONTRADICTION   | $a_s \Rightarrow \neg a_t$                         | yes             | (1, 1)                    |
| EQUIVALENCE     | $a_s \Leftrightarrow a_t$                          | no              | (0, 1) and (1, 0)         |
| EXCLUSIVE       | <i>exactly one of <math>a_s, a_t</math> holds</i>  | no              | (0, 0) and (1, 1)         |
| CO_NECESSITY    | <i>at least one of <math>a_s, a_t</math> holds</i> | no              | (0, 0)                    |
| NONE            | <i>no coupling</i>                                 | —               | none (no factor)          |

The first three are the NLI edge types of §3, so a mined relation round-trips into the existing machinery unchanged; the last two are additions required by argumentative prose. EXCLUSIVE is strictly stronger than CONTRADICTION, forbidding the both-false world as well — what *exhaustive* alternatives assert (“no one was harmed” versus “three people died” can neither both hold nor both fail) — and CO\_NECESSITY is its dual. We call  $\{\text{CONTRADICTION}, \text{EXCLUSIVE}\}$  the *conflict* couplings, being those that down-weight the both-true world.

#### 4.3 THE PAIRWISE POTENTIALS

The with-priors tables are *row-stochastic* — each source-state row sums to one, giving  $\mathbb{P}(a_t=1 \mid a_s=0) = \pi_s$  and  $\mathbb{P}(a_t=1 \mid a_s=1) = p$  for ENTAILMENT — and this is the section’s substantive modelling choice. Under the no-priors table the (1, 0) cell charges the *source* being true whenever the target is uncertain: appropriate when the source is a retrieved context whose own truth is at stake, wrong for coherence, where a well-supported premise must not be penalized for being a premise. The consequence is not cosmetic: on the example of §10 the spine’s root collapses to  $\mathbb{P}(a_1=1) = 0.19$  and the mean marginal stays at 0.46 *even after every conflict is removed*.

#### 4.4 LEVEL 2: DISCOURSE SENSES AND THE COMPILE MAP

Couplings are the right vocabulary for a factor table and the wrong one for annotation: asked whether two claims stand in ENTAILMENT, an annotator has been given a question about logic when the observable fact is one about discourse. We therefore mine an interpretable layer and compile it down.

**Definition 3** (Compile map). *Let  $\mathcal{S}$  be the set of Level-2 discourse senses. The compile map  $\mathcal{C} : \mathcal{S} \rightarrow \mathcal{T} \times ([0, 1] \cup \{\perp\}) \times \{0, 1\}^3$  assigns to each sense a coupling, a strength prior ( $\perp$  = use the estimator’s own value), and the flags directed, concession and ordering-only. It is given in Table 2 and is total and deterministic.*

The Contrast/Alternative boundary is the distinction the taxonomy exists to draw, and it is exactly the exhaustiveness question: claims that merely cannot both hold are a CONTRADICTION, those that additionally cannot both fail are EXCLUSIVE. A measure conflating them mis-scores every adjudicated dispute, since the point of a finding is to close an exhaustive pair.

**Concession discount.** A Concession is a conflict the text itself resolves — “the airline argued pilot error, but the report found a metallurgical defect” is not an incoherence if the report’s finding is asserted as authoritative. When a resolving *holding* atom  $h$  is present, the conflict’s weight is discounted,

$$p_r^{\text{eff}} = p_r (1 - \lambda \pi_h), \quad (3)$$

Table 2: The compile map  $\mathcal{C}$  (Def. 3). *Ordering-only* senses record assertion order but induce **no factor**, which is what makes an order-only edit provably score-invariant. **Concession** is a conflict the text may itself resolve (Eq. 3). The nine senses marked  $\star$  are the admissible vocabulary of §4.5.

| Level-2 sense        | $\rightarrow$ coupling | prior   | dir. | flags | gloss                          |
|----------------------|------------------------|---------|------|-------|--------------------------------|
| Cause-Effect $\star$ | ENTAILMENT             | $\perp$ | yes  |       | source causes target           |
| Effect-Cause $\star$ | ENTAILMENT             | $\perp$ | yes  |       | source is the effect           |
| Evidence $\star$     | ENTAILMENT             | $\perp$ | yes  |       | source is evidence for target  |
| Condition            | ENTAILMENT             | $\perp$ | yes  |       | source conditions target       |
| Instantiation        | ENTAILMENT             | $\perp$ | yes  |       | target instantiates source     |
| Restatement $\star$  | EQUIVALENCE            | 0.90    | no   |       | definitional identity          |
| Contrast $\star$     | CONTRADICTION          | $\perp$ | no   |       | opposed, <i>not</i> exhaustive |
| Concession $\star$   | CONTRADICTION          | $\perp$ | yes  | conc. | a resolvable tension           |
| Alternative $\star$  | EXCLUSIVE              | $\perp$ | no   |       | exhaustive: exactly one        |
| Disjunction $\star$  | CO_NECESSITY           | $\perp$ | no   |       | at least one                   |
| Precedence $\star$   | NONE                   | —       | yes  | ord.  | temporal order only            |
| Succession           | NONE                   | —       | yes  | ord.  | temporal order only            |

with  $\lambda = 0.45$ ; absent a resolver the relation stands at full strength. Holdings are identified by cue phrases (*held, ruled, concluded, the tribunal*) on an atom that is not itself an endpoint.

#### 4.5 ESTIMATING RELATIONS FROM THE RESPONSE

The relation weight decomposes into a type term and a strength term,

$$p_r = \underbrace{\mathbb{P}(\tau \mid a_s, a_t, y)}_{\text{type confidence}} \cdot \underbrace{\mathbb{P}(a_t \mid a_s, \tau, y)}_{\text{conditional strength}}, \quad (4)$$

estimated by two prompts. **Prompt A** elicits the discourse sense and coupling, with chain-of-thought *before* the label so the type confidence reads off the label span’s token logprobs exactly as in Eq. 1. **Prompt B** elicits the conditional strength as a surrogate Yes/No token, renormalized as  $\mathbb{P}(\text{Yes}) / (\mathbb{P}(\text{Yes}) + \mathbb{P}(\text{No}))$  (Kadavath et al., 2022); a verbalized alternative is kept only for comparison, verbalized confidence being weakly calibrated (Xiong et al., 2024).

**Grounding is mandatory.** Prompt A always sees the response and asserts a coupling only if the response *itself* draws the connection. Judging a pair in isolation makes a model accept any abstractly plausible relation, over-connecting the graph and — the score being a product of many soft constraints — collapsing the marginals toward the prior even for a coherent response. Two further precision mechanisms: an *admissibility filter* restricting output to the nine (sense, coupling) combinations marked  $\star$  in Table 2, and a required *verifiable evidence span* — a verbatim quotation from the response, checked by the parser rather than merely requested.

**Candidate pairs.** Mining all  $n(n-1)$  ordered pairs is quadratic and, worse, over-connects. We define four policies — ALL\_PAIRS; WINDOWED (forward pairs with  $0 < j - i \leq w$ ,  $w = 4$ ); GATED (the window plus long-range forward pairs passing a similarity gate); and BIDIRECTIONAL (both arc directions within  $|j - i| \leq w$ ) — and refine all of them against the response, *promoting* out-of-window pairs the prose links by shared entity or sentence adjacency and *demoting* in-window pairs it does not link (Appendix C). BIDIRECTIONAL exists because a forward-only policy *cannot emit* a directed arc running backward in atom order, so such gold relations are unreachable under it regardless of the model — a structural cap we measure in §9.

## 5 FOUR COHERENCE READOUTS

The MRF defines a distribution; a score is a functional of it. We give four candidates, then argue for one.

**(a) Mean marginal.** The headline readout,

$$\text{LCS}_{\text{mm}}(y) = \frac{1}{n} \sum_{i=1}^n \mathbb{P}(a_i = 1), \quad (5)$$

read directly off the marginals: the expected fraction of the response’s atoms that the argument, taken as a whole, upholds. A conflict suppresses its endpoints’ marginals; a satisfied entailment chain propagates support along itself.

**(b) Consistency.** A two-term readout separating the absence of conflict from the presence of support:

$$\text{LCS}_{\text{cn}} = \frac{1}{2}(\mathcal{K} + \mathcal{S}), \quad \mathcal{K} = \mathbb{P}\left(\bigwedge_{r: \tau_r = \text{CON}} \neg(a_{s_r} \wedge a_{t_r})\right), \quad \mathcal{S} = \frac{\sum_{r \in \mathcal{R}_{\mathcal{S}}} p_r U_r}{\sum_{r \in \mathcal{R}_{\mathcal{S}}} p_r}, \quad (6)$$

where  $\mathcal{R}_{\mathcal{S}}$  ranges over  $\{\text{ENTAILMENT}, \text{EQUIVALENCE}, \text{EXCLUSIVE}\}$  and  $U_r$  is the probability that  $r$  is *actively upheld*:  $\mathbb{P}(a_s=1 \wedge a_t=1)$  for entailment and equivalence,  $\mathbb{P}(a_s \neq a_t)$  for exclusive.

The two coupling sets differ deliberately.  $\mathcal{K}$  keys on CONTRADICTION alone: an EXCLUSIVE spreads its incoherence over *both* same-value worlds, so reading only its both-true half describes half the coupling and reports the other half as fine. It earns credit in  $\mathcal{S}$  instead, where its exactly-one world is both where the exclusion is honoured and where it is informative. CO\_NECESSITY enters neither term: its defect is the both-false world, which (a)’s marginals already see.  $\mathcal{K}$  is computed exactly with deterministic auxiliary variables — one  $u_r = a_{s_r} \wedge a_{t_r}$  per conflict edge, then a chained disjunction  $U = \bigvee_r u_r$ , and  $\mathcal{K} = \mathbb{P}(U=0)$  — all at most ternary, so no  $2^{|\mathcal{R}|}$  table is built.  $\mathcal{S}$  needs its own pass, being a weighted sum of *pairwise joints*: a linear functional of the joint, not an event probability.

**Proposition 1** (Why upholding, not satisfaction). *Let  $\mathcal{S}'$  credit a relation when it is merely satisfied rather than actively upheld, and let  $\mathcal{K}' = \mathbb{P}(\text{all relations satisfied})$ . Then a response with no mined relations receives  $\mathcal{K}' = 1$ , the maximum.*

The proposition is immediate — an empty product is one — but its consequence is not. The obvious repair to  $\mathcal{K}$ , extending the conflict event to “all relations honoured”, makes matters strictly worse: on the family of §10 a disconnected bag of atoms scores 1.000 while a response with a fully satisfied causal spine scores 0.221, converting a tie into an inversion, and every count-normalized variant we tried (geometric mean, expected fraction satisfied) is either still 1.000 on the bag or non-monotone across the ladder. The root cause is that “are the mined relations honoured?” is a property of the *relation set*, not of the response: it is maximized by asserting nothing. Crediting only the informative world (1, 1) removes the free lunch, since a bag of atoms upholds nothing and scores  $\mathcal{S} = 0$ . The terms are averaged, not multiplied, so neither can zero out the other.

**(c) Reified coherence node.** Add a binary variable  $R$  with prior  $\rho$  and one ternary vote factor per relation, charging  $1 - p_r$  in the  $R=1$  branch when  $r$  is violated at  $(a_s, a_t)$  and flat in the  $R=0$  branch; then  $\text{LCS}_{\text{re}} = \mathbb{P}(R=1)$  (Appendix A). *Violated* is the penalized set of Def. 2, so ENTAILMENT counts as satisfied even vacuously at  $a_s = 0$ .

**(d) Normalized log-partition.** The partition function measures how much probability mass the constraints leave, graded between two references from the same edge skeleton:

$$\text{LCS}_{\text{lp}} = \frac{\log Z - \log Z_{\min}}{\log Z_{\max} - \log Z_{\min}}. \quad (7)$$

$Z_{\max}$  removes all conflict edges — dropping constraint factors only adds mass, so  $\log Z \leq \log Z_{\max}$ . The floor is the *MAP world mass* of the base network,  $Z_{\min} = \max_{\mathbf{a}} \prod_i \phi_i \prod_r \psi_r$ .

**Lemma 1.**  $\log Z_{\min} \leq \log Z$  for any network, and  $Z_{\min}$  is coherence-graded.

*Proof.*  $Z = \sum_{\mathbf{a}} \text{mass}(\mathbf{a})$  is a sum of non-negative terms, so the single largest term bounds it below. Strengthening a conflict lowers the mass of every world that violates it, including the most probable one, so the bound moves with coherence.  $\square$

Lemma 1 is not a technicality: two natural skeleton-derived floors — retyping every edge to CONTRADICTION, or saturating conflicts to  $p = 1$  — are *not* valid lower bounds for row-stochastic tables, because an isolated conflict removes no mass while a mass-concentrating entailment backbone concentrates a great deal. Empirically the base log  $Z$  fell below both on real graphs, putting Eq. 7 outside  $[0, 1]$ .

### 5.1 WHICH READOUT, AND WHY

We judge the four on five criteria: **R1** a functional of the joint, not of the relation list; **R2** bounded in  $[0, 1]$  and interpretable; **R3** monotone — resolving a conflict never lowers the score; **R4** no hand-tuned constants; **R5** computable from standard inference queries. Only (a) and (d) satisfy all five (Table 5). Readouts (b) and (c) fail R3 for one shared reason worth stating precisely, because it looks like a bug and is not.

**Proposition 2** (Belief versus event activity). *Softening a resolved conflict via Eq. 3 raises the probability of its both-true world. For a lone conflict edge with uniform priors, discounting  $p$  from 0.80 to 0.55 moves  $\mathbb{P}(a_s=1, a_t=1)$  from 0.10 to 0.225.*

The both-true cell is  $1 - p$  for both conflict couplings, so weakening the constraint makes the forbidden world more probable. A readout asking *is the conflict active* — (b) counts that event, (c) charges  $1 - p_r$  — therefore registers a softened conflict as *more* often active and dips, while one asking *how strongly do we believe each atom* — (a) via marginals, (d) via mass — sees less suppression and rises. Resolving a concession should raise coherence, so a coherence score must track belief, not event activity. We select (a): it meets all five criteria, needs one MAR query, and is the only readout attributable to individual atoms, hence diagnosable. We report all four, since (b) and (d) carry complementary information ( $\mathcal{K}$  and  $\mathcal{S}$  separately, and the mass reading) and (b)’s dip is a *predicted* inversion that §7 turns into a testable constraint.

**Inference cost.** All four readouts need seven inference calls (base MAR and PR, two consistency passes, reified MAR, ceiling PR, base MAP); computed independently they would cost fourteen, each repeating the shared base pair. Production inference is weighted mini-bucket (Liu & Ihler, 2011) with i-bound 6; every *exact* value in §10 comes from  $2^n$  enumeration, our regression oracle for  $n \leq 20$ .

## 6 COMPOSING COHERENCE WITH FACTUALITY

The unary priors of Definition 1 are a free slot, and the natural thing to put in them is what Stage 1 already computed (Figure 1):

$$\pi_i \leftarrow q_i = \mathbb{P}(a_i = 1 \mid \mathcal{C}). \quad (8)$$

An atom is then credited when it is both externally supported and internally upheld, while setting  $q_i \equiv \frac{1}{2}$  recovers a pure coherence score consulting no evidence — so the two axes stay separable and the composition is an option, not a commitment. Two stages rather than one joint network is deliberate: inference runs over  $n$  atom variables rather than  $n(1+k)$ , the coherence graph can be rescored under several prior sets without re-mining, and the factuality stage stays independently cacheable. Priors are applied at scoring time, never by mutating the mined graph.

Matching Stage-1 atoms to Stage-2 atoms is done by object identity first, then normalized text, and only then by id — text before id deliberately, because near-duplicate removal leaves the surviving id set sparse, so matching on id alone can attach a prior to the *wrong* atom (Appendix B).

## 7 LOCoBENCH

Validating a coherence measure requires responses that differ in coherence while holding content fixed. Natural corpora do not supply this: two documents differing in coherence almost always differ in topic, length and claim set too, so any score difference is unattributable, and human ratings give a target but not a controlled contrast. What is needed is a family of responses over *one* claim set whose relative ordering is known by construction and whose gold relation graph is known, so scoring error separates from extraction error.

Table 3: The four ladder types; each holds claim set, topic and unperturbed relations fixed, varying one dimension. CONTROL is the false-positive check: every edit is meaning-preserving and adds no factor, so any monotone trend there means the measure is tracking edit count, not coherence.

| Type     | Rungs (0 $\rightarrow$ 4)                                                  | Varies                       | Required pattern                       |
|----------|----------------------------------------------------------------------------|------------------------------|----------------------------------------|
| CONFLICT | worse, base, concession_resolved, fix_one_conflict, coherent               | conflict structure           | strictly increasing                    |
| CHAIN    | break_3, break_2, break_1, base, coherent                                  | chain integrity              | strictly increasing                    |
| ORDER    | shuffle_full, shuffle_block, shuffle_adjacent, base, ordering_only         | assertion order              | increasing, then <i>flat</i> at rung 4 |
| CONTROL  | ordering_only, direction_reversal, base, direction_reversal, ordering_only | nothing (meaning-preserving) | <b>flat</b>                            |

### 7.1 COHERENCE LADDERS

**Definition 4** (Coherence ladder). A ladder is a tuple  $(\mathcal{P}, \langle \rho_0, \dots, \rho_4 \rangle, \Gamma)$  where  $\mathcal{P}$  is a base relation plan — a claim set with a declared relation graph — each rung  $\rho_k$  is a sequence of perturbation operator calls applied to  $\mathcal{P}$  yielding a response and its own gold relation graph, and  $\Gamma$  is a set of ordering constraints over the rungs’ scores. A ladder is admitted only if all five rungs pass every validation gate.

All-or-nothing admission is forced by the design: four of the five rungs carry no standalone claim, only one relative to their neighbours, so the ladder — not the item — is the unit of validity.

Ordering constraints come in three classes, asserted per readout: **C1**, strict increase between consecutive rungs; **C2**, a *predicted* decrease or invariance — where Proposition 2’s concession dip becomes a falsifiable prediction rather than an excuse, and where an *ordering-only* rung must be exactly flat since Table 2 gives it no factor; and **C3**, endpoint separation between rungs 0 and 4. A measure is scored on the fraction it satisfies.

### 7.2 GENERATION

Generation is a five-stage LLM pipeline with fixed, versioned prompts.

- P1** *Topic  $\rightarrow$  question*. One open-ended question whose answer will naturally carry diverse logical structure. The prompt asks for adjudicative domains — incident investigations, litigation, competing hypotheses — which produce exhaustive alternatives and resolving holdings, and forbids the words *cause*, *condition*, *alternative*, *contrast* so relations are not primed lexically.
- P2** *Question  $\rightarrow$  claims*. 26 tagged atomic claims: 14 correct, 4 plausible-but-false, an *exhaustive alternative pair* of which exactly one must hold, a disjunctive pair, and two *holding* claims able to resolve a tension.
- P3** *Claims  $\rightarrow$  relation plan*. Select 14–16 claims, renumber to consecutive assertion positions, and plan 8–12 relations over distinct pairs, each with a sense from Table 2, a validity label and a strength band. At least two false claims must be selected, so factuality cannot be inferred from coherence.
- P4** *Plan  $\rightarrow$  response*. Prose realizing the plan exactly: assert each atom once, in position order, referring back by pronoun rather than restatement. Quantifiers, numbers, dates, polarity and modals are content, not wording.
- P5** *Response  $\rightarrow$  rung*. Apply one rung’s perturbation calls, changing the minimum number of words (operators in Appendix F).

### 7.3 VALIDATION

Every threshold lives in one table and generation is rejected rather than repaired. An independent *committee*, distinct from the generator, runs three majority-gated validators. **V1** is load-bearing: recover the relations the response *itself* draws, admitting only if  $\geq 0.80$  of the *valid* planned relations come back with the correct coupling and  $\geq 0.70$  with the correct sense — this certifies that the prose



realizes the plan, so the gold graph describes the response and not the plan’s intention. **V3** audits the prose (fluency, formality, organization each  $\geq 4$  of 5; no span may reveal the text came from an annotation plan) and **V4** requires every atom present. Plan structure is gated separately, including a  $0.55 \pm 0.15$  valid fraction and  $\geq 2$  false claims selected, so factuality cannot be inferred from coherence. Appendix E gives every threshold and two measurement subtleties: the 4–6 *declared non-relations* per item, the only real negative set against which a false positive is meaningful, and the separation of relation *identity* (the unordered pair) from the *grading* of direction, coupling and sense.

**Per-rung gold relations.** An earlier generator perturbed each rung’s *text* but re-attached the base plan’s relation graph to all five rungs. The gold relations were then constant within a family, so every readout was constant too and the strict-increase constraints could not hold by construction: the gold arm satisfied 2/28. Deriving each rung’s edges from its own operator calls, and rejecting any family where two *adjacent* rungs build the same edge set while the higher rung’s calls claim a change, raises this to 22/28 on the same topics and generator. This is the clearest evidence that the ladder tests a measure rather than the corpus — and a warning that a benchmark of this shape can silently be untestable.

## 8 EVALUATION PROTOCOL AND BASELINES

An *arm* fixes where the relation graph comes from, holding atoms, priors and readouts identical: GOLD takes each item’s own labelled relations, GOLD\_VALID only those labelled valid, and a MINED arm runs the estimator of §4.5 on the response text. This is the central experimental device — the gold arm is a ceiling isolating *scoring* error, so any shortfall below it is *extraction* error. Mined relations are compared to gold at three nested levels: **pair** (were these two atoms related at all), **coupling** (pair, plus the Level-1 coupling, with direction required for asymmetric couplings), and **sense**. The coupling level is the one that matters, being the level at which the factor table is chosen.

A coherence measure should be compared against controls chosen to fail informatively. We propose five families (Appendix G): *ladder-blind controls* (atom count, relation count, length, perturbation-call count), which must be flat on CONTROL and fail C3 elsewhere, and which catch the likeliest way such a measure is secretly wrong; *local contradiction detection*, pairwise NLI with no global model (Manakul et al., 2023), isolating what joint inference adds; *LLM judges* — G-Eval’s coherence dimension (Liu et al., 2023) and a direct 1–5 rating — the practical incumbent, and a demanding test since rungs differ by a few words; *discourse representations*, DiscoScore (Zhao et al., 2023) and an entity grid (Barzilay & Lapata, 2008), strongest on ORDER and weakest on CONFLICT; and *internal ablations* over priors, readouts, pair policies, prompt variants and the gold ceiling. Beyond ladder ordering we recommend Spearman correlation against human ratings: ordering agreement is the sharper instrument, being a controlled contrast, but human correlation establishes that the construct ordered is the one readers care about.

## 9 EXPERIMENTS

**Scale and sampling, stated first.** The corpus reported here is **two CONFLICT families, ten items**, produced by `claude-opus-5` with a three-model committee (`gpt-5.6`, `gemini-3.1-pro`, `claude-sonnet-5`) running V1/V3/V4; relations are mined by `llama-3.3-70b-instruct`. The CHAIN, ORDER and CONTROL ladders are implemented and gate-checked but *not yet generated*, so no result below speaks to them. Mining is sampled once per cell at default temperature, and repeated runs of an identical configuration moved coupling-level F1 by up to 0.06; across five samples the two revised strategies span 0.49–0.58, every one above the unrevised 0.36, so the gap to the original exceeds the noise while the ordering *between* the revisions does not. These are measurements, not significance claims; the second decimal of any single cell should not be read.

**Relation extraction is the bottleneck.** Three findings from Table 4. ALL\_PAIRS achieves the best recall (82.6%) and by far the worst precision (9.3%): mining every pair finds the true relations and buries them, emitting 867 edges against 86 gold, and since the readouts are products over relations this is what drives the marginals toward the prior. The forward-only policies are *structurally* capped — backward recall 23–27% against 73.3% — because 20 of the 86 gold edges are both directed and backward and no forward-only candidate set can emit them; admitting both arc directions within the

Table 4: **Top:** mined-versus-gold agreement over 86 edge-producing gold relations, micro-averaged; **Fwd/Bwd** split recall by whether the gold edge runs forward or backward in atom order, and **Non-rel** counts violations of the declared non-relations (a real negative set). **Bottom:** ordering constraints satisfied (28 assertions per arm) and each readout averaged over items. GOLD\_VALID is near-flat by construction — a `fix_one_conflict` rung removes the planted-invalid conflicts it has already discarded — so its failures say nothing about the readouts; GOLD is what the constraints test.

| Policy / strategy           | Edges | Pair R | Coup P | Coup R | Coup F1     | Fwd  | Bwd  | Non-rel     |
|-----------------------------|-------|--------|--------|--------|-------------|------|------|-------------|
| WINDOWED                    | 190   | 83.7   | 25.8   | 57.0   | 35.5        | 75.0 | 23.3 | 7/48        |
| GATED                       | 321   | 83.7   | 14.6   | 54.7   | 23.1        | 69.6 | 26.7 | 8/48        |
| ALL_PAIRS                   | 867   | 95.3   | 9.3    | 82.6   | 16.8        | 87.5 | 73.3 | 10/48       |
| BIDIRECTIONAL               | 205   | 83.7   | 39.0   | 66.3   | 49.1        | 71.4 | 56.7 | 5/48        |
| BIDIRECTIONAL <sup>v2</sup> | 173   | 73.3   | 48.3   | 67.4   | <b>56.3</b> | 85.7 | 33.3 | <b>0/48</b> |

| Arm                         | f001  | f002  | %           | mm    | cn    | re    | lp    |
|-----------------------------|-------|-------|-------------|-------|-------|-------|-------|
| GOLD                        | 11/14 | 11/14 | <b>78.6</b> | 0.796 | 0.092 | 0.232 | 0.383 |
| GOLD_VALID                  | 7/14  | 4/14  | 39.3        | 0.793 | 0.157 | 0.292 | 0.462 |
| MINED, WINDOWED             | 4/14  | 9/14  | 46.4        | 0.843 | 0.007 | 0.360 | 0.164 |
| MINED, GATED                | 7/14  | 8/14  | 53.6        | 0.679 | 0.002 | 0.262 | 0.102 |
| MINED, ALL_PAIRS            | 7/14  | 5/14  | 42.9        | 0.590 | 0.163 | 0.037 | 0.018 |
| MINED, BIDIR.               | 5/14  | 5/14  | 35.7        | 0.810 | 0.000 | 0.403 | 0.143 |
| MINED, BIDIR. <sup>v2</sup> | 6/14  | 8/14  | 50.0        | 0.660 | 0.053 | 0.366 | 0.088 |

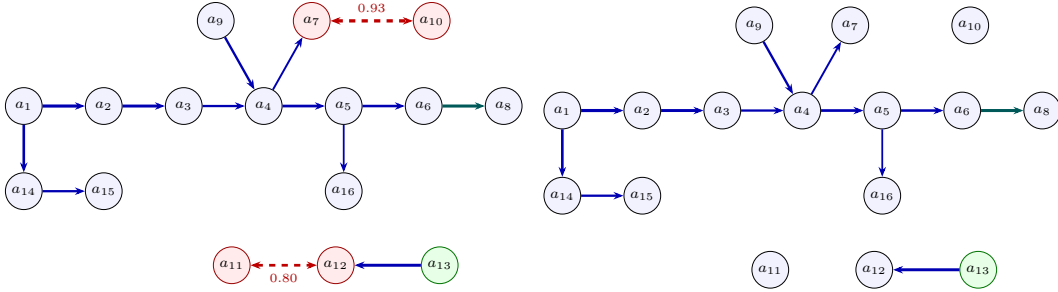
window recovers most of that gap at a fifth of the calls. And the two prompt revisions (rebalancing the few-shot label distribution toward the true base rate, and requiring a verifiable evidence span) raise coupling F1 from 35.5 to 56.3 while driving declared-non-relation violations to zero and illegal-vocabulary emissions from 25 to 0. Precision, not recall, is where a grounded miner is won.

**The model orders correctly when the graph is correct.** On gold relations the readouts satisfy 78.6% of the declared constraints (Table 4) — 22 of 28, against 2/28 pre-fix — while mined arms reach 35.7–53.6%. Since readouts, atoms and priors are identical across arms, the 25–43 point gap is attributable to extraction, consistent with Table 4: at 56.3% coupling F1 roughly half the factors in a mined network are wrong or missing, and a ladder whose rungs differ by one or two edges cannot survive that. Arm ordering does not track F1 — GATED scores best among mined arms at the second-worst F1 — which at 28 assertions per arm we read as noise, and which is why the corpus must grow before policies are ranked on ladder agreement.

**Readout dynamic range is policy-dependent.** On nine hand-authored fixtures scored across three models and three strength estimators, the readouts differ sharply in spread. Under the response-grounded WINDOWED policy the estimator asserts almost no contradiction edges, so  $cn \equiv 1.00$  and  $re \equiv 0.50$  (spread 0.00) while  $mm$  (spread 0.87) and  $lp$  (spread 1.00) discriminate, separating a consistent biography ( $mm = 0.99$ ) from the same biography with five planted contradictions (0.82); under ALL\_PAIRS the same readouts regain range ( $re$  spans 0.00–0.50). The degeneracy is thus a property of the *graph*, not the readout:  $\mathcal{K} = 1.00$  is the *correct* answer for a conflict-free response, verified on an authored post-mortem ( $mm = 0.939$ ,  $\mathcal{K} = 1.000$ , 1.9 relations per atom), and is uninformative exactly when the estimator asserts no conflicts at all. Aggregated over all cells, contradiction-heavy fixtures average  $mm = 0.560$  against 0.767 for conflict-free ones. Hence reporting  $mm$  as headline with  $\mathcal{K}$ ,  $\mathcal{S}$  as diagnostics: a number that is 1.00 both for a sound argument and an unmined one is not usable alone.

## 10 TWO WORKED EXAMPLES

The response is a 16-atom aviation recall report. Its relations are 12 support edges — an entailment spine  $a_1 \rightarrow \dots \rightarrow a_6$  (shipment, installation, service entry, in-flight failures, probe, grounding order) with branches, plus one EQUIVALENCE — and two conflicts. All values are exact, from  $2^{16}$  enumeration.



(a) **Incoherent** (base). Two EXCLUSIVE conflicts (dashed):  $a_7 \leftrightarrow a_{10}$  is *unresolved*;  $a_{11} \leftrightarrow a_{12}$  is a concession the holding  $a_{13}$  (green) resolves.  $\text{mm} = 0.607$ ,  $\log Z = -9.64$ . (b) **Coherent**. The identical support structure with both conflict edges removed;  $a_{10}$  and  $a_7$  are no longer suppressed.  $\text{mm} = 0.620$ ,  $\log Z = -8.25$ ,  $\text{lp} = 1.00$ .

Figure 2: Relation graphs for the AeroParts recall report (§10). Solid blue: ENTAILMENT; teal: EQUIVALENCE; dashed red, double-headed: EXCLUSIVE. Line width grows with the relation weight  $p$ . The two panels differ only in the two conflict edges, so the  $0.607 \rightarrow 0.620$  lift and the 1.39 nats of recovered mass are attributable to conflict structure alone.

**Example 1** (Incoherent). *The report asserts both that no passengers or crew were harmed and that a wire report claimed three passengers died. These are exhaustive alternatives, not mere opposition: they cannot both hold and cannot both fail, so the relation is EXCLUSIVE with  $p = 0.93$ , and nothing in the response resolves it. A second tension — the airline’s pilot-error argument against the report’s metallurgical-defect finding,  $p = 0.80$  — is a **Concession**, resolved by the tribunal holding  $a_{13}$  and discounted by Eq. 3. The model returns  $\text{LCS}_{\text{mm}} = 0.607$  and  $\log Z = -9.64$  (Figure 2a); the marginal of the atom on the losing side of the unresolved conflict is dragged below its own prior, which is the per-atom diagnostic that localizes the incoherence. Applying the concession discount alone raises the score to 0.612.*

**Example 2** (Coherent). *Removing the two conflict edges and changing nothing else leaves the identical support structure (Figure 2b). The score rises to  $\text{LCS}_{\text{mm}} = 0.620$  with  $\log Z = -8.25$  and  $\text{LCS}_{\text{lp}} = 1.00$ : the response is now as coherent as its own edge skeleton allows, which is what the ceiling of Eq. 7 means. The suppressed atom recovers to its prior.*

Because the panels differ *only* in the two conflict edges, the  $0.607 \rightarrow 0.620$  lift and the 1.39 nats of recovered mass are attributable to conflict structure alone — the property the ladder construction of §7 generalizes. The full CONFLICT ladder is monotone throughout:  $0.586 < 0.607 < 0.612 < 0.613 < 0.620$  over the rungs worse, base, concession\_resolved, fix\_one\_conflict, coherent.<sup>1</sup>

A second fixture pair makes the order-sensitivity claim concretely: two summaries of one appellate judgment from the *same* facts, differing only in whether self-serving claims precede or follow the findings that qualify them. The faithful ordering scores  $\text{mm} = 0.95$ , the self-serving-first ordering 0.90 — and no per-atom check can separate them, every atom being identical.

## 11 LIMITATIONS

The empirical scope is the main limitation and is stated in §9: two families, ten items, one CONFLICT ladder type, and single-sample mining with measured run-to-run movement of 0.06 F1. The CHAIN, ORDER and CONTROL ladders are specified and gate-checked but ungenerated, so the false-positive check CONTROL provides — arguably the most important single experiment in the protocol — has not been run, and the baselines of §8 are argued for rather than measured.

Three modelling limitations. The vocabulary is pairwise by construction, so genuinely  $k$ -ary structure (three claims jointly inconsistent, none pairwise so) is outside the family; the transitivity and double-conflict rules that would capture it are expressible only in the Markov logic view of Appendix H.

<sup>1</sup>Treating both tensions as plain CONTRADICTION rather than EXCLUSIVE — the three-coupling model of Marinescu et al. (2025) — gives 0.587 and  $\log Z = -9.75$  for Example 1, with Example 2 unchanged at 0.620 and  $-8.25$ . The exhaustiveness distinction of §4 is worth 0.020 here.

The concession discount’s  $\lambda = 0.45$  and the holding-cue heuristic are the one place hand-set constants enter. And production inference is approximate while the exact values reported come from enumeration ( $n \leq 20$ ); the agreement of the two is not characterized at length. Finally the corpus is LLM-generated throughout, gold labels included, with LLM-committee validation: V1’s  $\geq 0.80$  gate certifies the prose realizes the plan, but a systematic bias shared between generator and committee would escape it, so human verification of a sample is required before treating the corpus as ground truth.

## 12 CONCLUSION

Coherence is a property of a joint distribution over what a response asserts, and it can be read off a Markov random field whose factors are the relations the response itself draws. Formalizing the vocabulary as two levels — interpretable discourse senses compiling deterministically onto five inferential couplings, each fixed by the world it penalizes — keeps annotation tractable while holding the model to a closed pairwise family. Of four readouts, the mean posterior marginal alone is monotone in conflict resolution, bounded, constant-free and per-atom diagnosable; two plausible alternatives fail for a reason worth remembering, that a score asking whether asserted relations are honoured is maximized by asserting nothing. Measuring such a score requires controlled contrasts, and LoCoBench supplies them as ladders whose orderings are declared before any measurement. Gold relations recover 78.6% of those orderings, mined relations 53.6%; since the arms differ only in the relation graph, the gap localizes the open problem — the probabilistic model is adequate and grounded relation extraction, at 56.3% coupling-level F1, is the bottleneck. That is a more tractable target than “measure coherence”, and it is measurable on a corpus we release.

### AI USE STATEMENT

We used generative AI tools as *objects of study* and as *components of the released artifacts*, and we disclose both. All LoCoBench corpus content — questions, atomic claims, relation plans, responses and per-rung perturbations — is LLM-generated by the pipeline of §7, and its validation labels are produced by an LLM committee; §11 states the consequences for treating the corpus as ground truth. The relation estimator of §4.5 is LLM-based by design. We additionally used AI assistants for code implementation and for editorial assistance on this manuscript. All LLM-generated code was reviewed and is covered by the test suite that pins the numerical values reported in §10; every number in §9 was traced to a generated experiment record. The research questions, the taxonomy, the readout definitions and the selection arguments are the authors’ own. We take responsibility for the final content of this work.

### ETHICS STATEMENT

This work involves no human subjects and no personally identifying data. The fixtures include a summary of a published appellate judgment, used as a worked example of ordering-dependent coherence; it is a matter of public record and no private individual is the subject of analysis. A coherence measure can be misread as a truth measure: a response can be perfectly coherent and entirely false, which is exactly why §6 keeps the two axes separable, and we caution against deploying LCS alone as a factuality filter. The corpus deliberately contains false claims, labelled as such, and should not be redistributed without those labels.

### REPRODUCIBILITY STATEMENT

The model is fully specified in the paper: Definition 1 gives the joint, Table 1 the potentials, Table 2 the compile map, and Eqs. 5–7 the four readouts with their auxiliary constructions. Table 3 and Table 6 specify the generator, and §7 lists every acceptance threshold. The exact values in §10 are computed by  $2^{16}$  enumeration and are pinned by regression tests in the released code, so they are reproducible without LLM access; the relation graph they use is given edge-by-edge in Figure 2. §9 names the generator, committee and mining models, and states the sampling regime and its measured variance. The generator, the evaluation harness, the nine fixtures and the generated corpus are released with the code.

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Table 5: The four readouts against five criteria. **R1** a functional of the joint, not of the relation list; **R2** bounded in  $[0, 1]$  and interpretable; **R3** monotone — resolving a conflict never lowers the score; **R4** no hand-tuned constants; **R5** computable from standard inference queries.

| Readout           | Query          | R1 | R2 | R3 | R4 | R5 | Note               |
|-------------------|----------------|----|----|----|----|----|--------------------|
| (a) mean marginal | MAR            | ✓  | ✓  | ✓  | ✓  | ✓  | selected           |
| (b) consistency   | MAR $\times 2$ | ✓  | ✓  | ×  | ✓  | ✓  | concession dip     |
| (c) reified       | MAR            | ✓  | ✓  | ×  | ×  | ✓  | free $\rho$ ; dip  |
| (d) norm. log $Z$ | PR, MAP        | ✓  | ✓  | ✓  | ✓  | ✓  | needs 2 references |

## A THE REIFIED COHERENCE NODE

**(c) Reified coherence node.** Add a binary variable  $R$  with prior  $\rho$  and one ternary vote factor per relation,

$$h_r(R, a_s, a_t) = \begin{cases} 1 & R = 0 \\ 1 & R = 1, r \text{ satisfied at } (a_s, a_t) \\ 1 - p_r & R = 1, r \text{ violated,} \end{cases} \quad \text{LCS}_{\text{re}} = \mathbb{P}(R = 1). \quad (9)$$

*Satisfied* is the complement of the penalized set of Def. 2, so ENTAILMENT is satisfied everywhere but  $(1, 0)$  — including vacuously at  $a_s = 0$ .

## B ATOM ALIGNMENT ACROSS STAGES

**Alignment.** Matching Stage-1 atoms to Stage-2 atoms is done by object identity first (the reuse path, when atomization is shared), then by normalized text, and only then by id. Text precedes id deliberately: near-duplicate removal keeps the first-seen atom and leaves the surviving id set sparse, so two independent atomizations can disagree about which claim lives at which index, and matching on id alone would attach a prior to the *wrong* atom — worse than attaching none. When coverage falls below one half the run is flagged degraded, and the configured policy is to warn, raise, or discard every prior and fall back to a clean coherence-only score rather than a half-primed mixture.

## C CANDIDATE-PAIR POLICIES

**Candidate pairs.** Mining all  $n(n - 1)$  ordered pairs is quadratic and, worse, over-connects. We define four policies: ALL\_PAIRS; WINDOWED, forward pairs with  $0 < j - i \leq w$  ( $w = 4$ ); GATED, the window plus long-range forward pairs passing a similarity gate; and BIDIRECTIONAL, both arc directions within  $|j - i| \leq w$ . The last exists because a forward-only policy *cannot emit* a directed arc that runs backward in atom order, so such gold relations are unreachable under it regardless of the model — a structural cap we measure in §9. All policies are refined against the response: pairs outside the window that the prose links by shared entity or sentence adjacency are *promoted*, and in-window pairs the prose does not link are *demoted*.

## D READOUT SELECTION CRITERIA

## E VALIDATION GATES

Every acceptance threshold lives in one table, and generation is rejected rather than repaired. An independent *committee* of models — distinct from the generator — runs three validators, each gating by majority:

- **V1, relation recovery.** Given the response and its atoms, recover the relations the response *itself* draws. Admission requires  $\geq 0.80$  of the *valid* planned relations recovered with the correct coupling and  $\geq 0.70$  with the correct sense. This is the load-bearing gate: it certifies that the prose actually realizes the plan, so the gold graph describes the response rather than the plan’s intention.

Table 6: The six perturbation operators and their ten parameterized calls. Each call also has a declared effect on the rung’s *gold relation graph*, so a rung’s edges are the base edges with its own perturbations applied. O6’s calls are factor-invariant by design — they are what the ORDER and CONTROL ladders exist to check.

| Op               | Calls                                                       | Effect                                                                                                                             |
|------------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| O1 relabel       | wrong_sense,<br>direction_rev.,<br>exhaustive-<br>ness_flip | rewrite the connective so the text draws a different sense, reverses the relation, or flips Contrast $\leftrightarrow$ Alternative |
| O2 add conflict  | inject_contradiction,<br>spurious_relation                  | add a sentence contradicting a named atom, or assert a link the plan does not contain                                              |
| O3 break chain   | break_chain                                                 | remove or negate a link so the causal chain no longer connects, keeping both endpoints                                             |
| O4 drop relation | drop_relation                                               | delete one relation                                                                                                                |
| O5 resolve       | add_resolution,<br>remove_resolution                        | add or delete the holding sentence that resolves a conceded tension                                                                |
| O6 reorder       | shuffle_order,<br>ordering_only                             | reorder sentences, or swap Precedence $\leftrightarrow$ Succession: no factor changes                                              |

- **V3, prose audit.** Fluency, formality and organization each  $\geq 4$  of 5, with the *leakage* and *hedging* span lists empty — no span may reveal that the text was generated from an annotation plan.
- **V4, atom coverage.** Every atom present: coverage = 1.00 over the statuses *missing* and *merged*. The status *altered* is deliberately non-gating, being the judgment most sensitive to the surface rewording P4 licenses.

Plan structure is gated separately: 14–16 claims, 8–12 relations, 4–6 declared *non-relations*, a valid-relation fraction of  $0.55 \pm 0.15$ , and at least two false claims selected.

## F PERTURBATION OPERATORS

## G PROPOSED BASELINES IN DETAIL

1. *Ladder-blind controls*: atom count, relation count, response length, and number of perturbation calls applied. These must be flat on CONTROL and must fail C3 elsewhere. They cost nothing and they catch the most likely way a coherence measure is secretly wrong — by tracking edit count or graph density.
2. *Local contradiction detection*: pairwise NLI over atom pairs with no global model, scored as (one minus) the fraction of pairs labelled contradictory, in the spirit of SelfCheckGPT’s NLI variant (Manakul et al., 2023). This is the ablation that isolates what joint inference adds over local detection, and it should be expected to do reasonably on CONFLICT and poorly on CHAIN, where the defect is a *missing* entailment rather than a present contradiction.
3. *LLM judges*: G-Eval’s coherence dimension (Liu et al., 2023) and a direct “rate the logical coherence from 1 to 5” prompt. These are the practical incumbent and the comparison reviewers will want. The ladder is a demanding test for them because rungs differ by a few words, so a judge must be sensitive at very small edit distances while remaining flat on CONTROL.
4. *Discourse representations*: DiscoScore (Zhao et al., 2023), and an entity-grid model (Barzilay & Lapata, 2008) as the classical reference. Both are order-sensitive by construction, so ORDER is where they should be strongest and CONFLICT, where the perturbation is semantic rather than positional, where they should be weakest.
5. *Internal ablations*: uniform versus factuality priors (Eq. 8); the four readouts of §5; the four candidate-pair policies; the two prompt variants; and the gold-relation ceiling. These separate the model’s own design choices from the estimator’s accuracy.

## H CORRESPONDENCE WITH MARKOV LOGIC

The coherence MRF has a natural first-order encoding (Richardson & Domingos, 2006), with  $\mathbb{P}(\mathbf{a}) = \frac{1}{Z} \exp(\sum_k w_k n_k(\mathbf{a}))$  for  $n_k$  the number of true groundings of formula  $F_k$ . Evidence predicates  $Entail(i, j)$ ,  $Contradict(i, j)$ ,  $Equiv(i, j)$  and  $Resolves(h, i, j)$  carry the mined graph;  $Holds(i)$  is the query predicate.

The naive encoding — one weighted clause per relation, e.g.  $Entail(i, j) \wedge a_i \Rightarrow a_j$  with  $w = \text{logit}(p)$  — is **not** equivalent to Definition 1. Its factor  $[e^w, e^w, 1, e^w]$  is not row-stochastic: it rewards the vacuous  $a_s = 0$  rows on both target states. Brute force over  $2^{16}$  worlds gives mean marginal support 0.455 against the MRF’s 0.587, with per-atom marginals differing by up to 0.31 — two different models, not two encodings of one. Its mode is degenerate for the reason Proposition 1 identifies: the all-false world satisfies every implication vacuously, so the most coherent reading of any response is the empty one.

Expanding each relation into three clauses restores the correspondence. Writing  $\ln \psi(a_s, a_t) = \alpha + \beta a_s + \gamma a_t + \delta a_s a_t$ , the conjunction weight is exactly  $\pm \text{logit}(p)$  — the interaction the implication was meant to carry — and the source unit clause is the correction restoring row-stochasticity:

| $\tau$        | $\alpha$              | $\beta$             | $\delta$             |
|---------------|-----------------------|---------------------|----------------------|
| ENTAILMENT    | $\ln \frac{1-p}{p_s}$ | 0                   | $+\text{logit}(p)$   |
| CONTRADICTION | $\ln \frac{p}{p_s}$   | 0                   | $-\text{logit}(p)$   |
| EQUIVALENCE   | $\ln \frac{1-p}{p}$   | $\ln \frac{1-p}{p}$ | $+2 \text{logit}(p)$ |

With this expansion the ground Markov logic network reproduces the MRF exactly: mean support 0.5866 and  $\log Z = -9.751$  for both, with zero per-atom difference to machine precision. The expansion is established for the three original couplings; EXCLUSIVE and CO\_NECESSITY are not tabulated. The reason to prefer the MRF for the measure itself is that its pairwise family is closed at three parameters, so the vocabulary cannot quietly grow; the reason to keep the Markov logic view is that the beyond-pairwise rules (transitivity, concession cancellation, double conflict) are expressible there and are the natural route past the pairwise limitation noted in §11.